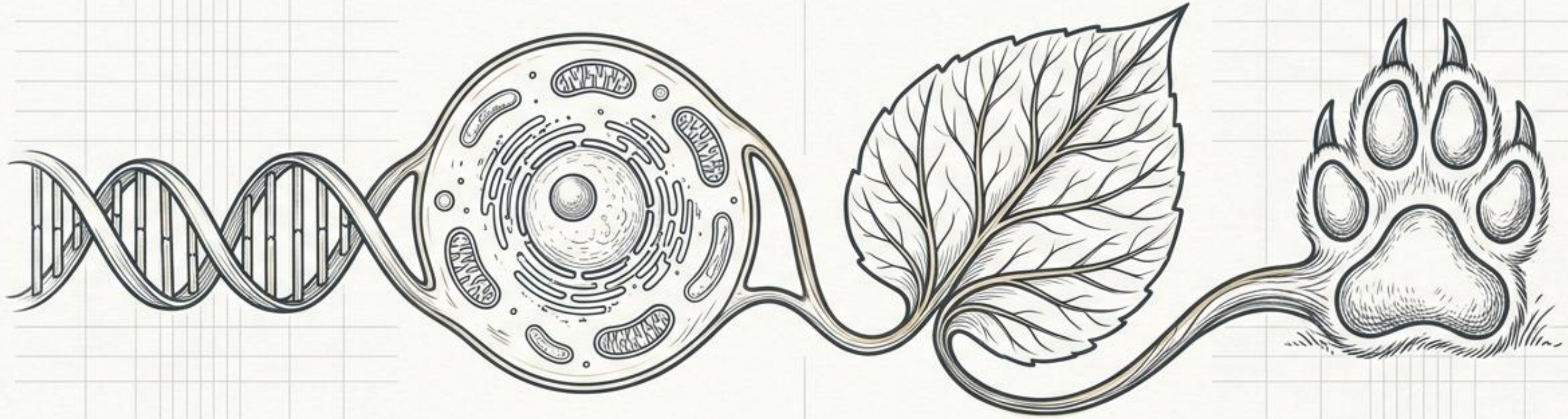


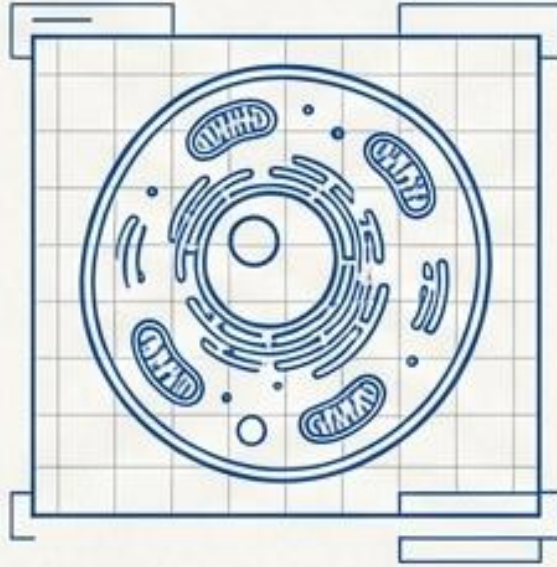


The Biological Blueprint: From Micro to Macro

A comprehensive visual study guide for Unit 2: The Organization of Life.

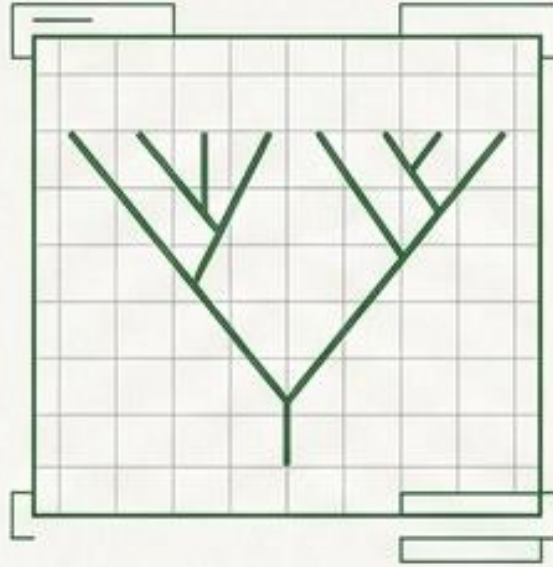
SUBJECT: Biology
EXAM FOCUS: Chapters 2-4
THEME: Scale & Structure

The Microscopic Foundation



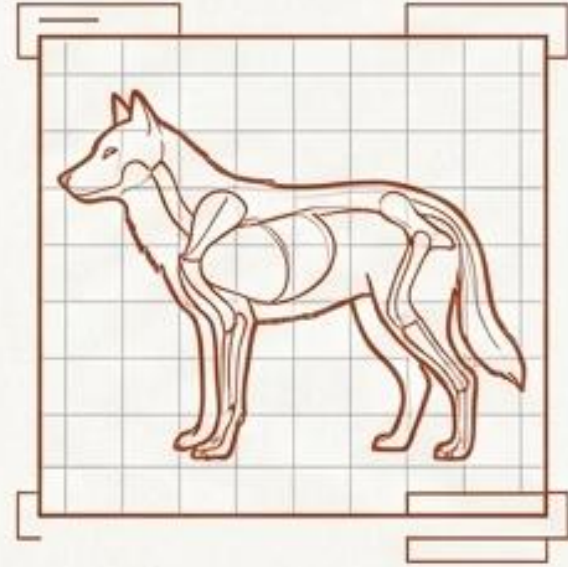
- Cellular Machinery
- Energy Economics
- Macromolecules

Ordering the Chaos



- Taxonomy
- Evolutionary Mapping
- Indigenous Frameworks

The Macro Machine

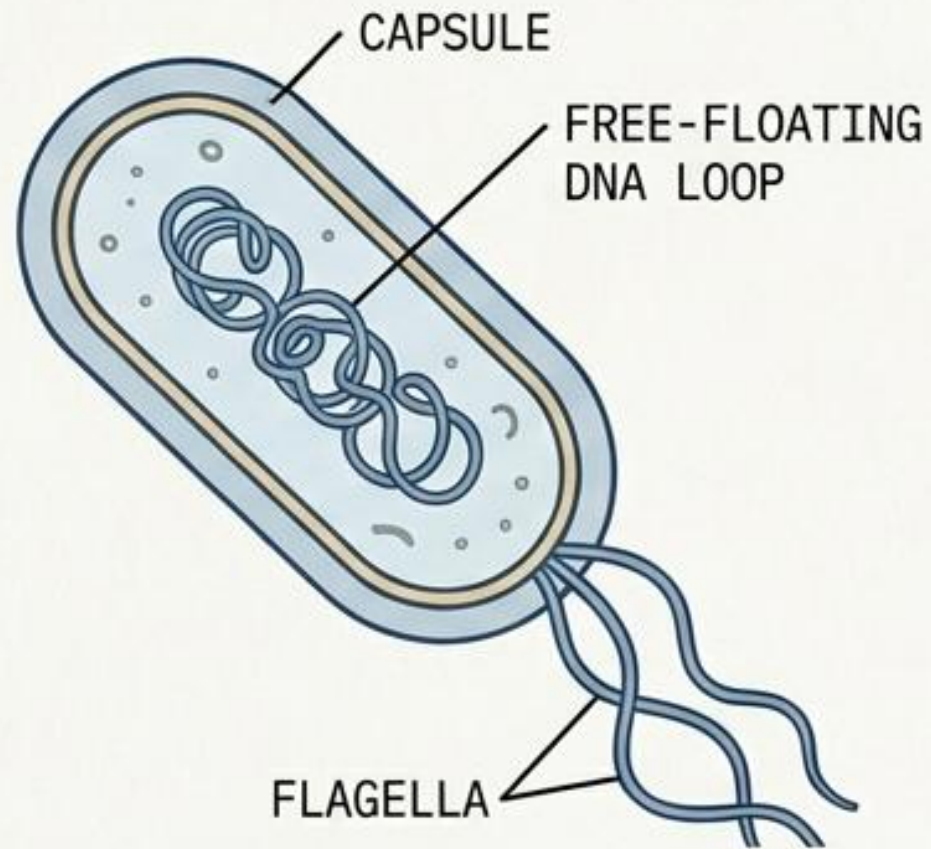


- Comparative Anatomy
- Homeostasis
- Survival Strategies

Life is built on architecture. Understanding biology means understanding how matter is organized at every scale.

The Two Empires of Life

Prokaryotes: The Ancient Simpletons

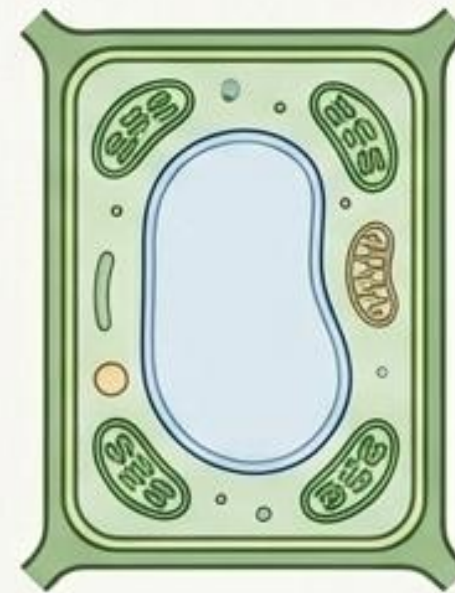


- No true nucleus (nucleoid region)
- Small scale (1-5 μm)
- Circular DNA loop
- Example: *Rhizobium* (Saskatchewan pulse crop soil bacteria)

Eukaryotes: The Complex Compartments

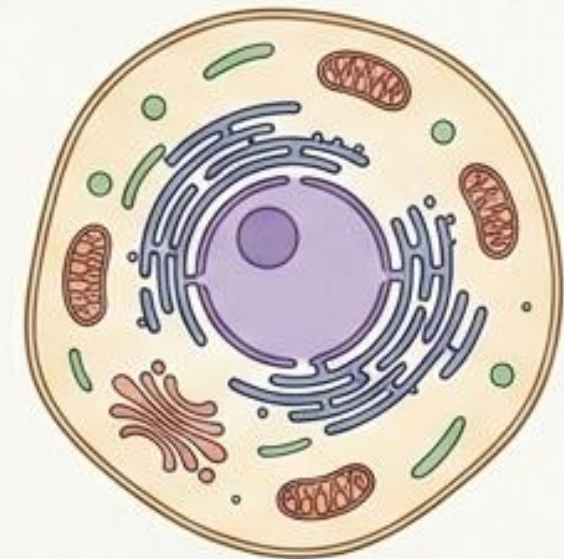
- True nucleus & linear chromosomes
- Large scale (10-100 μm)
- Membrane-bound organelles

Plant Cells



Chloroplasts, Cellulose
Cell Wall (tough fibers),
Large Central Vacuole.

Animal Cells



Centrioles, Lysosomes
(waste digestion),
Smaller vacuoles.

The Macromolecule Toolbox

Carbohydrates

Energy & Structure



- 1:2:1 C:H:O ratio.
- Monomer: Monosaccharides (Glucose).
- Polymer: Polysaccharides (Starch, Cellulose).

Lipids

Fluid Mosaic

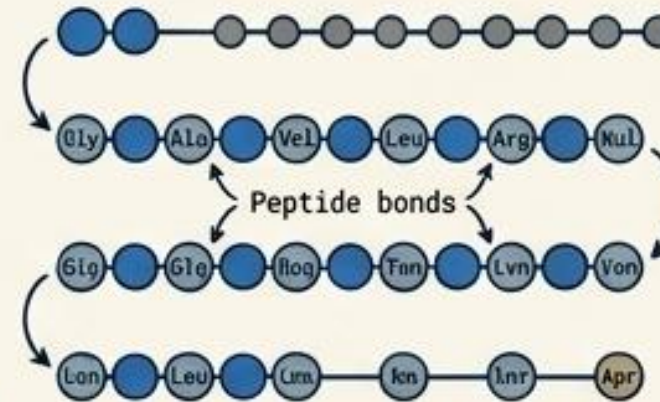


- Hydrophilic phosphate heads.
- Hydrophobic fatty acid tails.

Proteins

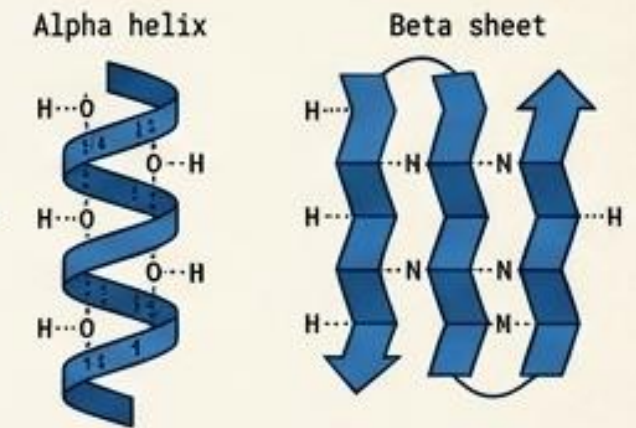
The Cellular Machinery

1. Primary Structure (Amino Acid Sequence)

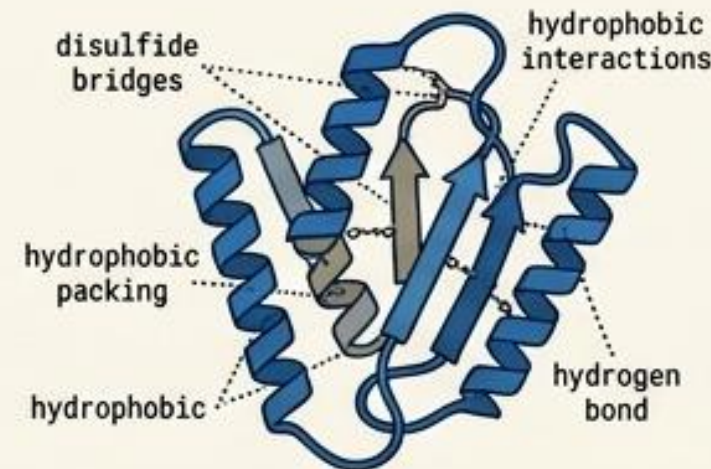


1. Primary (Sequence)

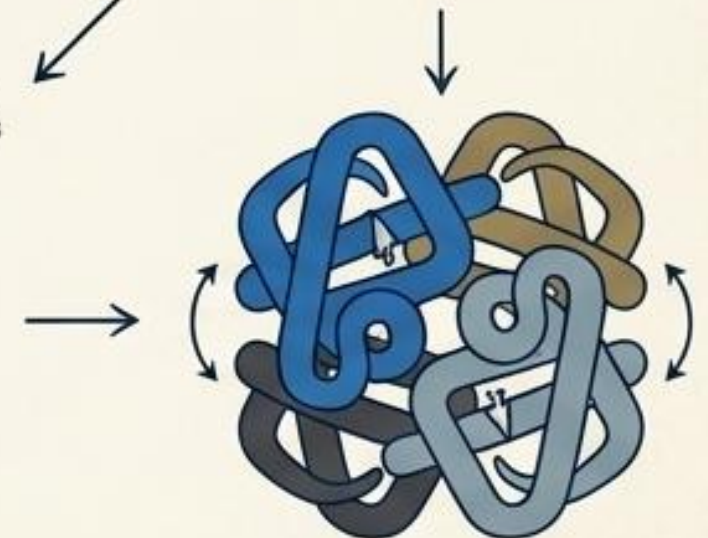
2. Secondary Structure (Alpha Helix & Beta Sheet)



-> 2. Secondary (Helix/Sheet)



-> 3. Tertiary (3D Fold)



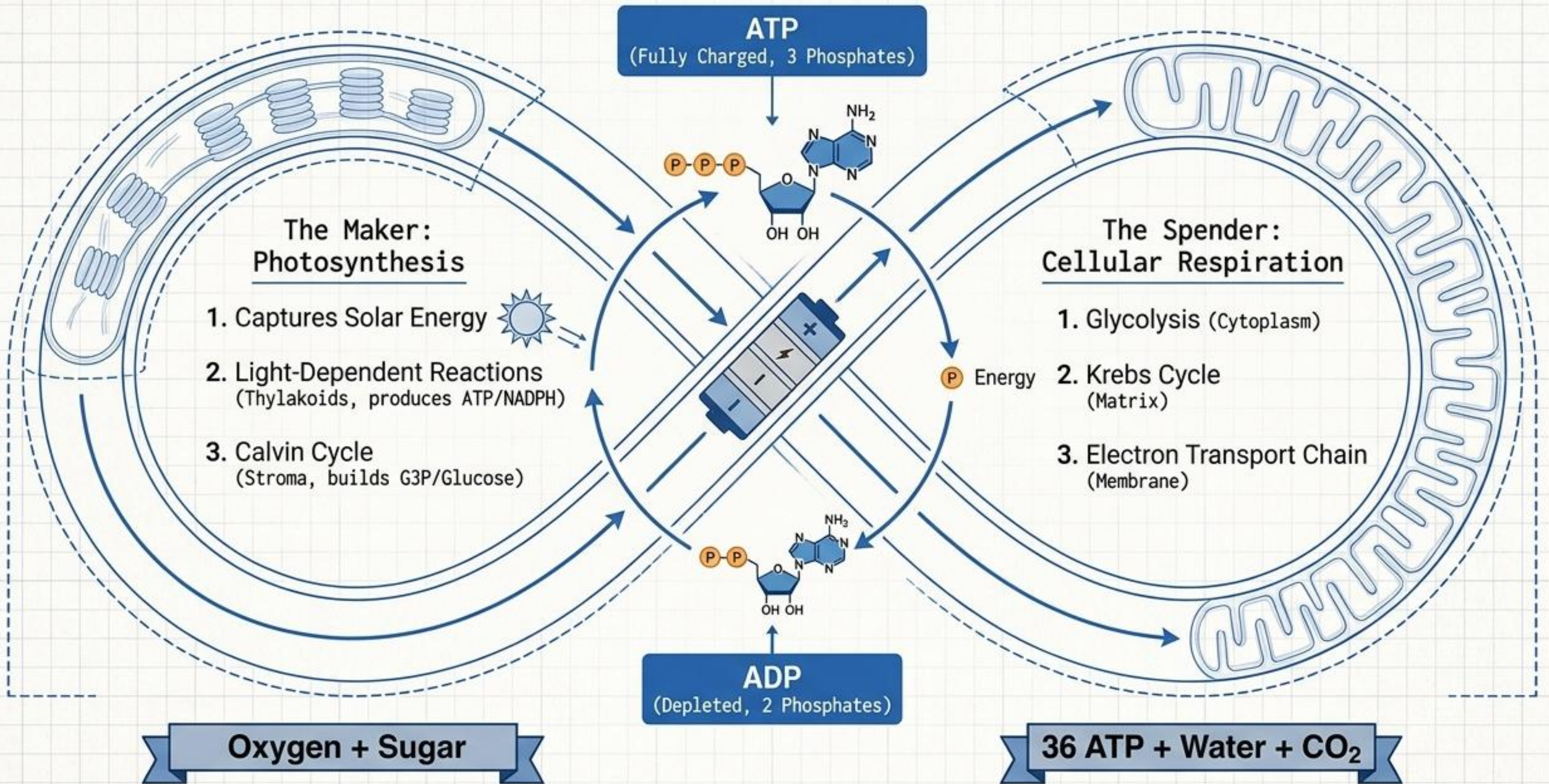
-> 4. Quaternary (Subunits)

Nucleic Acids

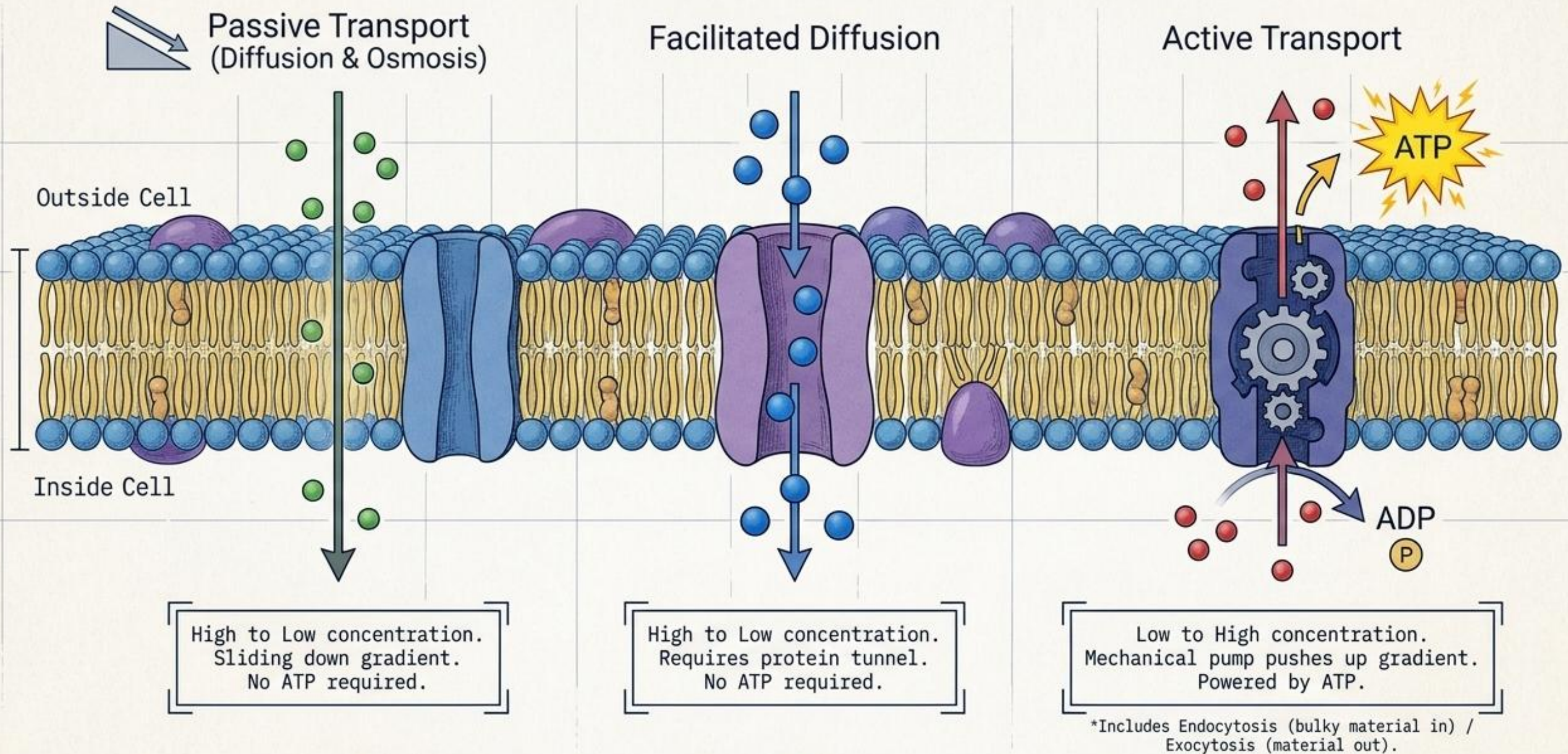
Information Storage



- DNA (Deoxyribose, A-T/C-G).
- RNA (Ribose, A-U/C-G).



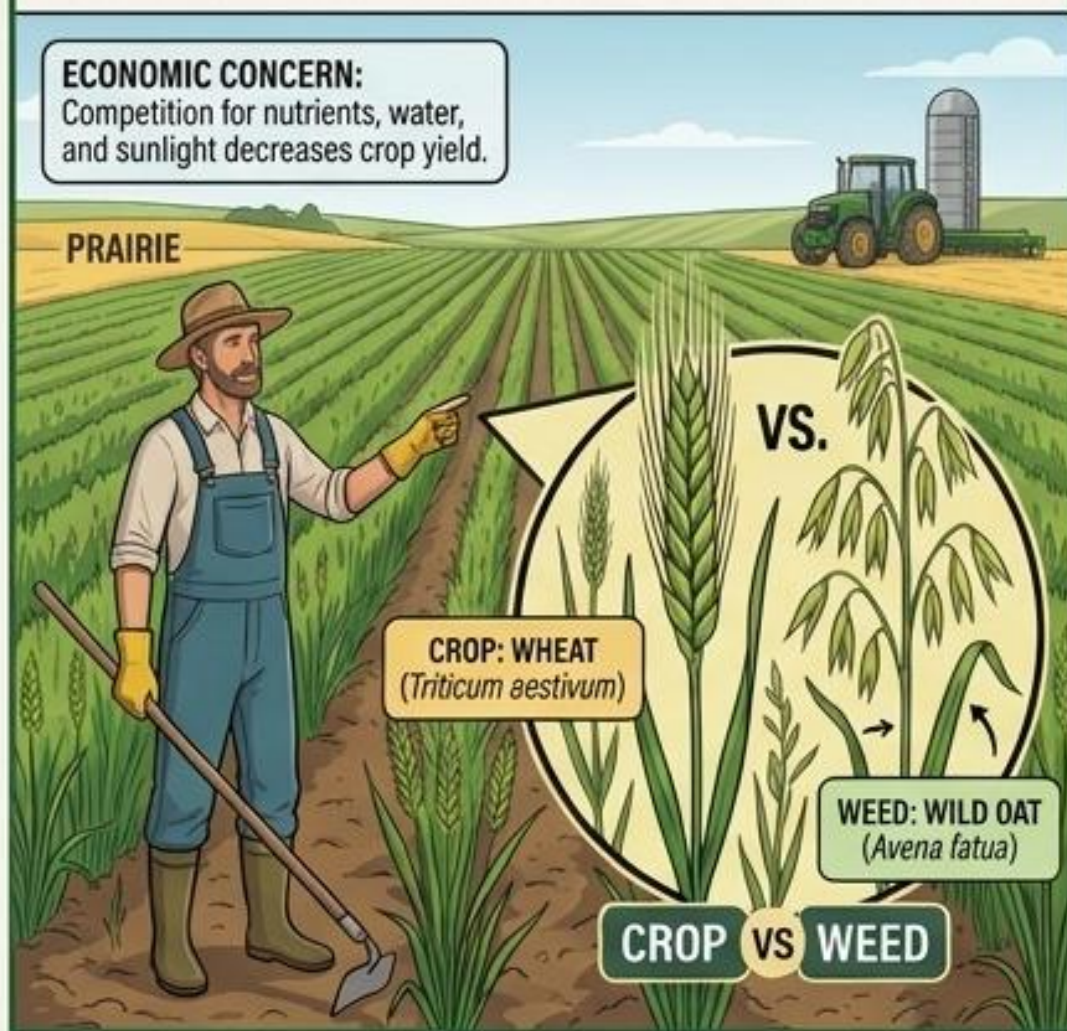
Crossing the Border: Membrane Transport



Organizing the Chaos: Frameworks of Classification

Artificial / Functional

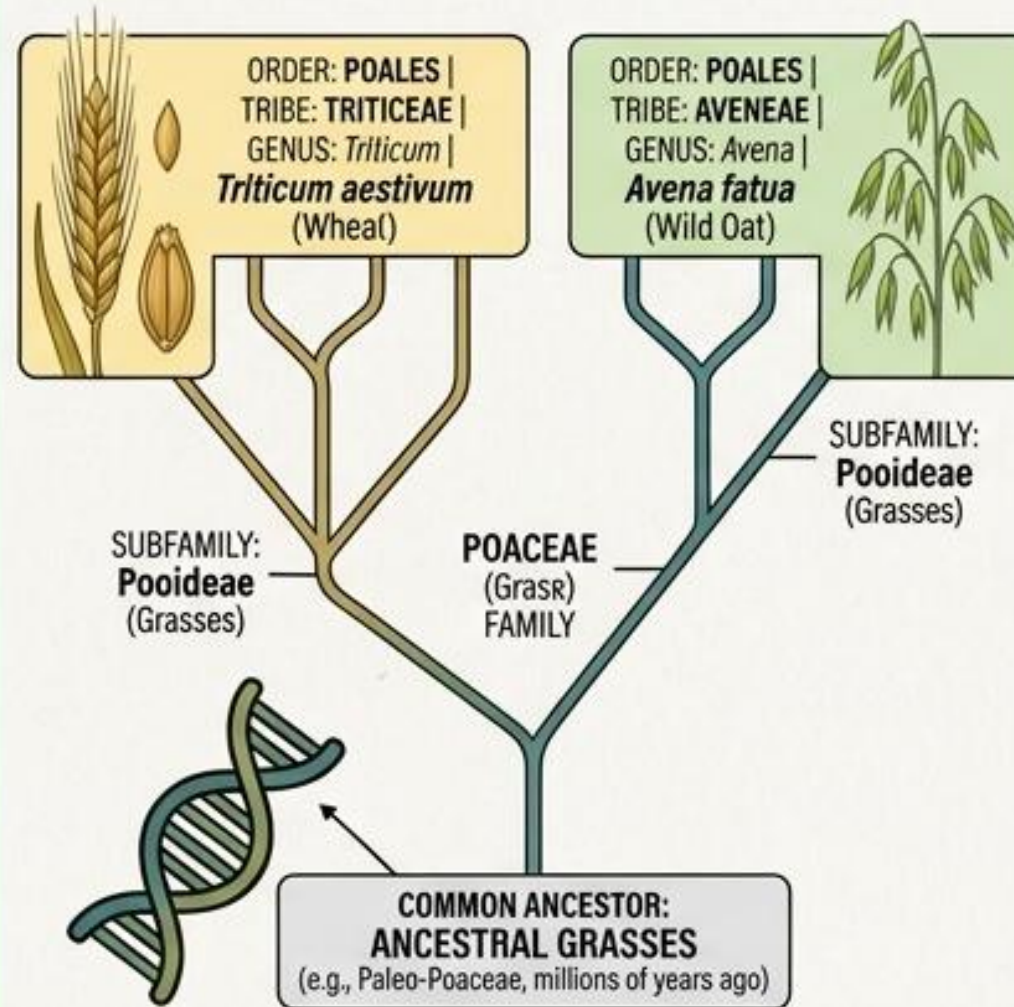
The Farmer's View



Groups by utility or arbitrary traits. Useful for human needs, but ignores biological reality (e.g., classifying by Crop vs. Weed).

Phylogenetic / Natural

The Biologist's View



Groups by deep-seated homologies and shared ancestry. Based on DNA and evolutionary history.

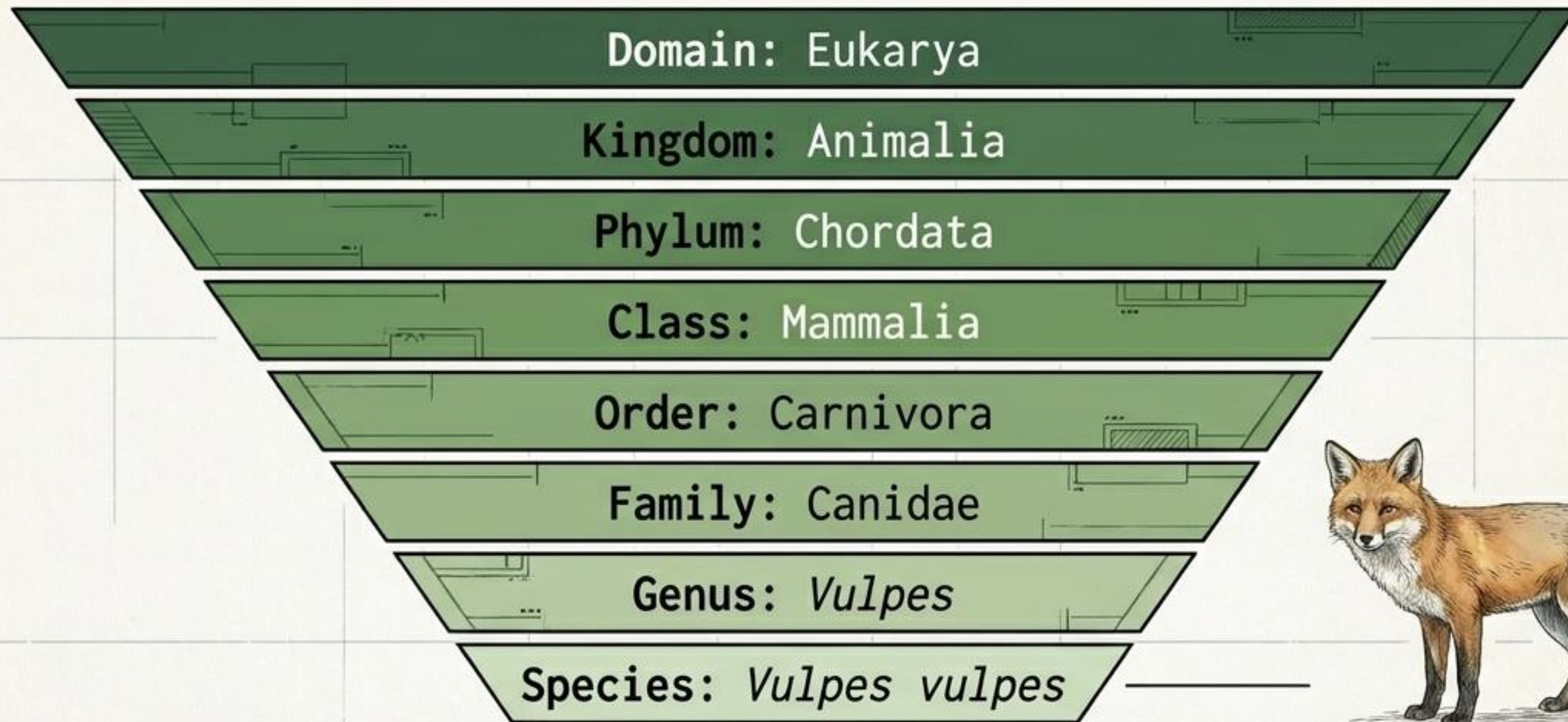
Wâhkôhtowin / Relational

The Indigenous View



Groups by kinship, balance, and interconnected cycles. Animals are relatives and teachers, emphasizing reciprocity rather than hierarchy.

The Taxonomic Ladder

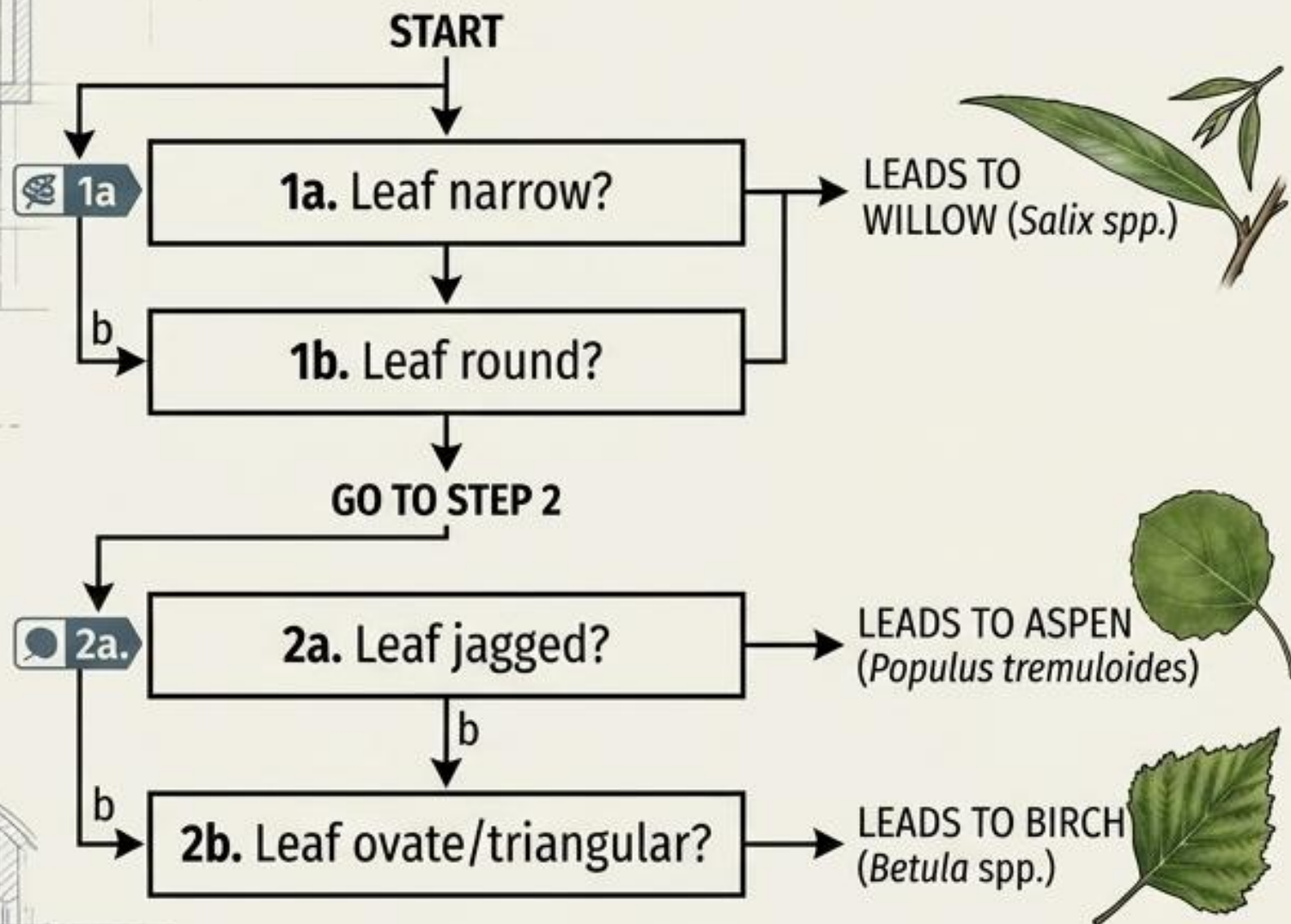


Binomial Nomenclature Rulebook

- Developed by Carl Linnaeus.
A two-part, stable Latin naming system.
- Rule 1: Genus is always capitalized.
• Rule 2: species is always lowercase.
- Rule 3: Always italicized or underlined.
• Example: *Homo sapiens*

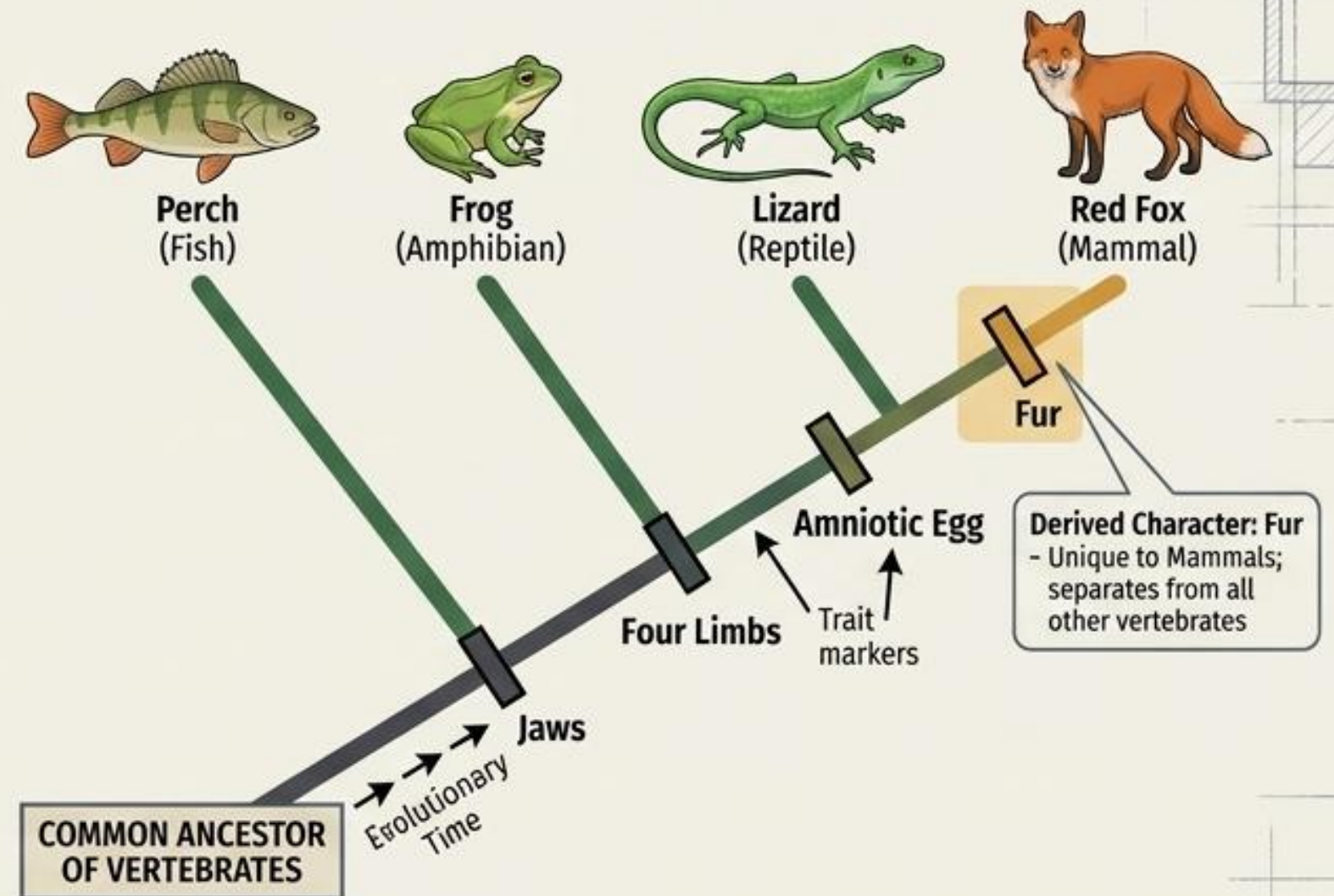
Mapping History: Dichotomous Keys & Cladograms

Identification: The Dichotomous Key



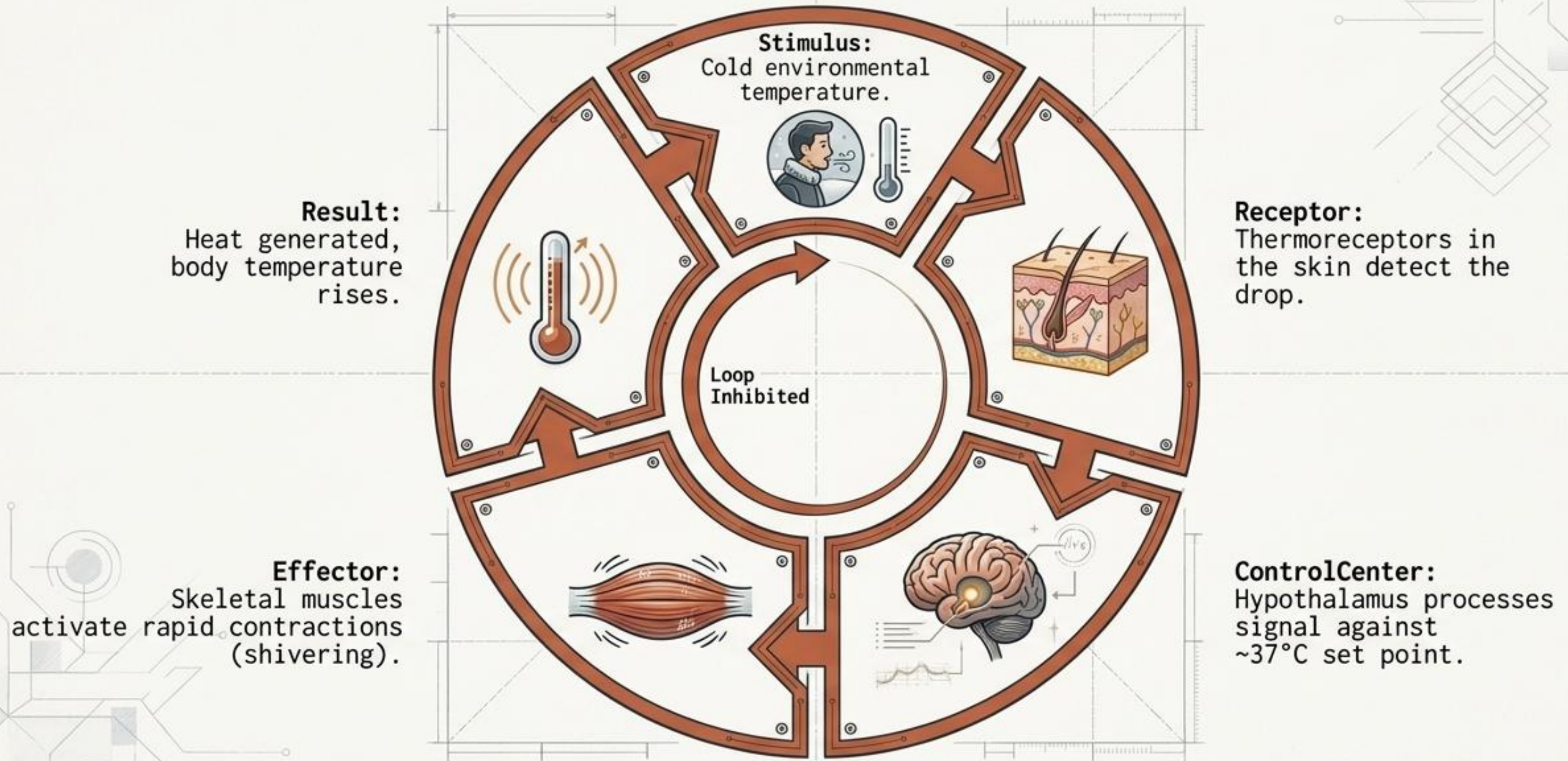
Choose Your Own Adventure for Identification based on binary physical choices.

History: The Cladogram



A timeline mapping derived characters and shared evolutionary ancestry.

The Homeostasis Engine: Negative Feedback



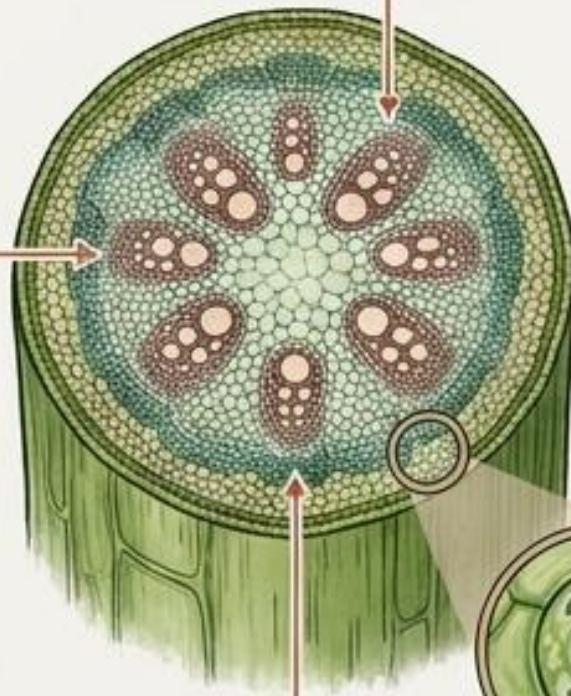
Negative Feedback acts to cancel out the original stimulus and restore the baseline set point.

NATURE'S CHANNELS: INTERNAL TRANSPORT SYSTEMS

Plantae: Anchored Autotrophs

XYLEM
Water/minerals
UP from roots.

PHLOEM
Sugars DOWN
from leaves.

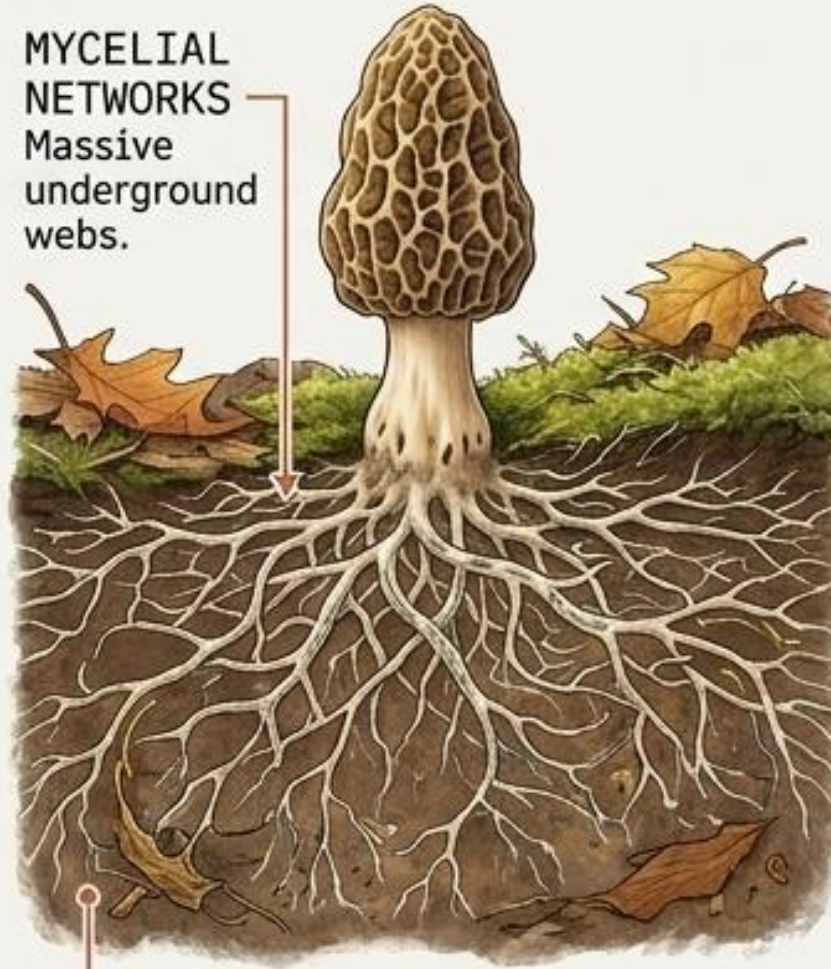


XYLEM
Water/minerals
UP from roots.

STOMATA
Pores for gas
exchange.

Fungi: Decomposers

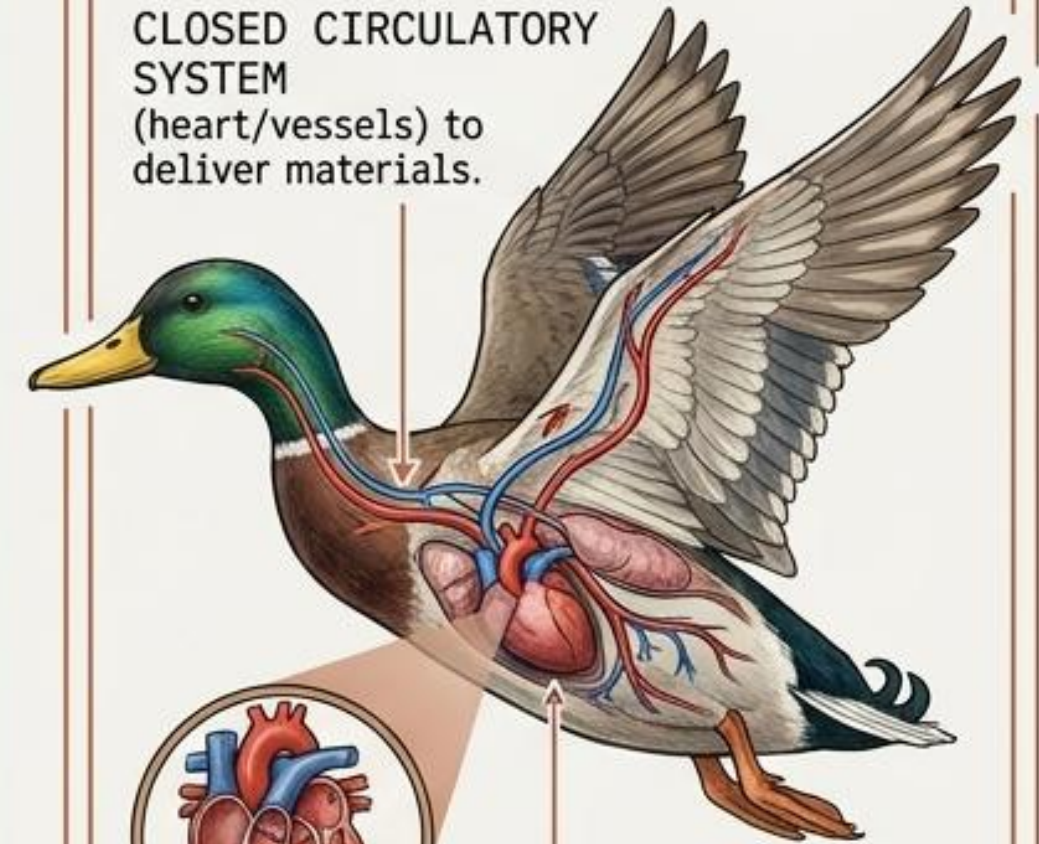
**MYCELIAL
NETWORKS**
Massive
underground
webs.



EXTRACELLULAR DIGESTION
Secretes enzymes outward to break
down food, then absorbs nutrients.

Animalia: Mobile Heterotrophs

**CLOSED CIRCULATORY
SYSTEM**
(heart/vessels) to
deliver materials.

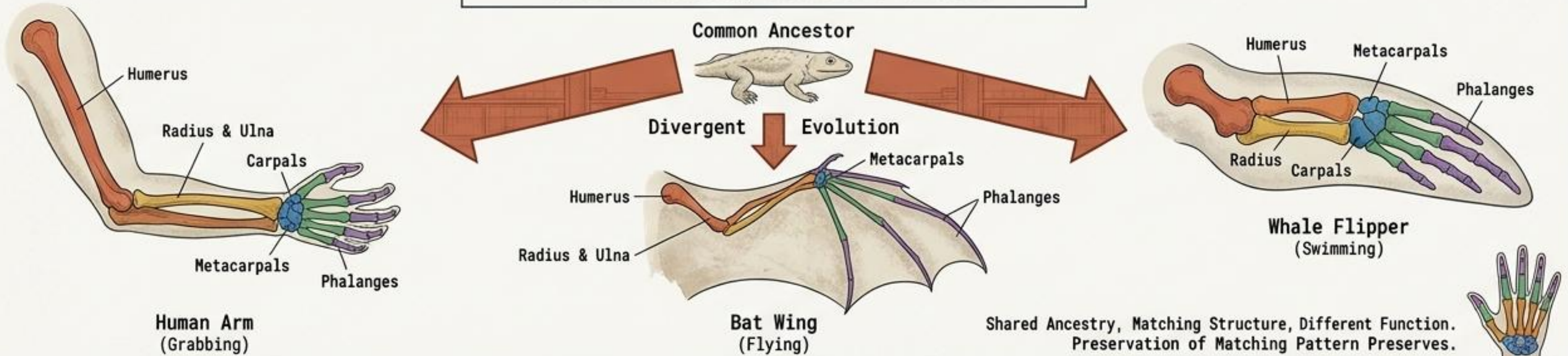


HEART
(Pump)

LUNGS/AIR SACS
for massive oxygen
acquisition.

HOMOLOGOUS STRUCTURES

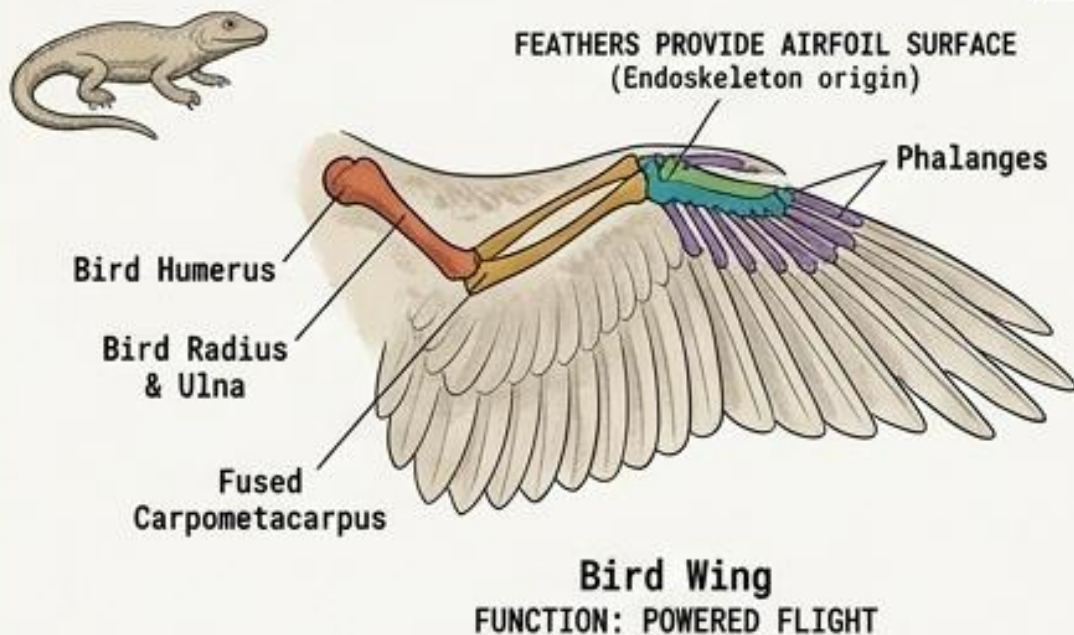
Shared Ancestry, Different Function



ANALOGOUS STRUCTURES

Shared Function, Different Ancestry

Different Ancestors

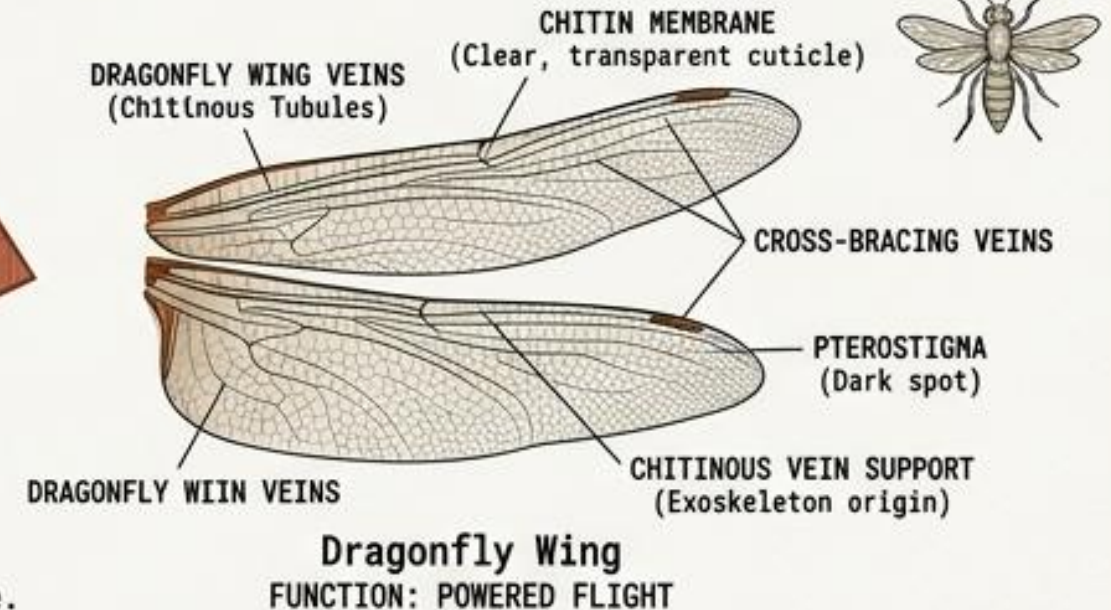


Convergent Evolution

Environmental Solution: Flight

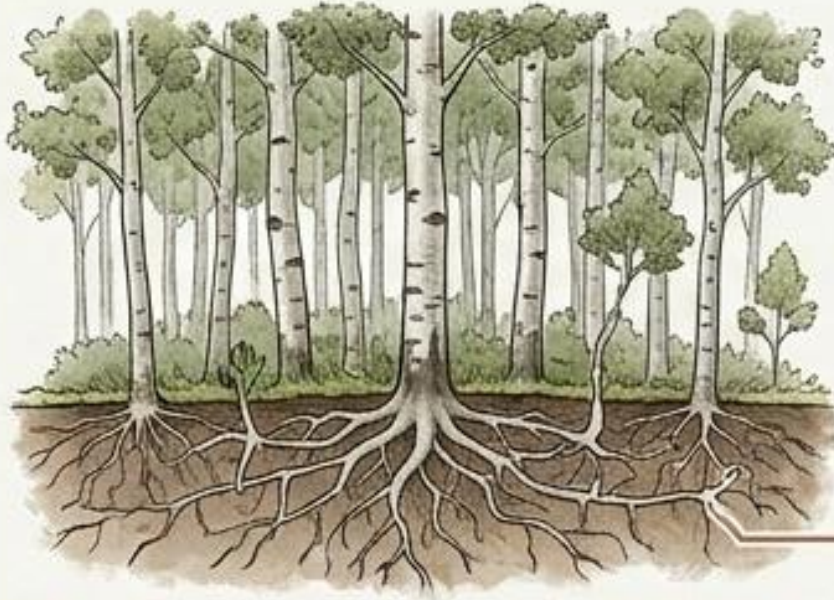
Different structural origins adapted to solve the identical environmental problem. Similar Function, Different Ancestry and Structure.

Different Ancestors



REPRODUCTIVE STRATEGIES: SURVIVAL TRADE-OFFS

Asexual Reproduction

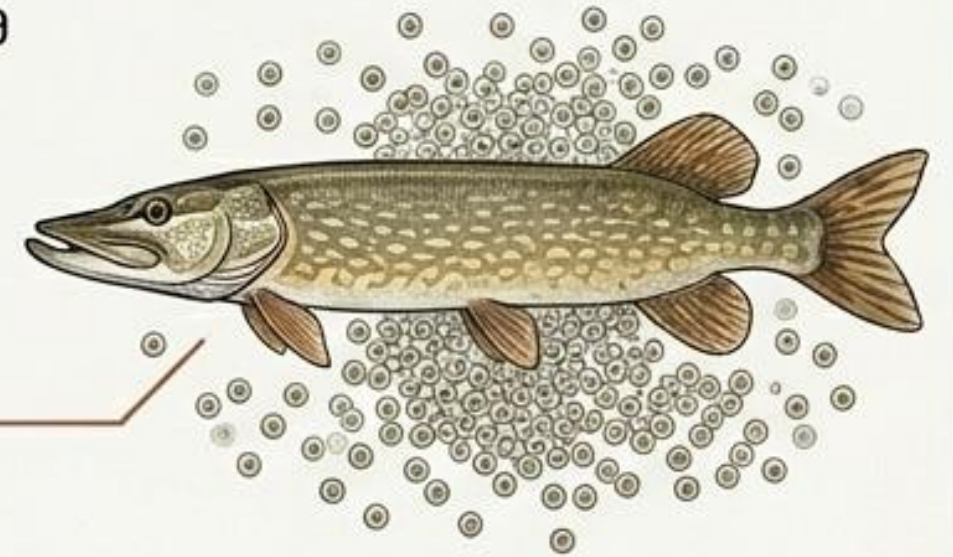


— 1 parent. Offspring are genetic clones. Extremely fast and efficient. Excellent for stable environments but high extinction risk from a single disease.

Trembling Aspen clonal colonies.

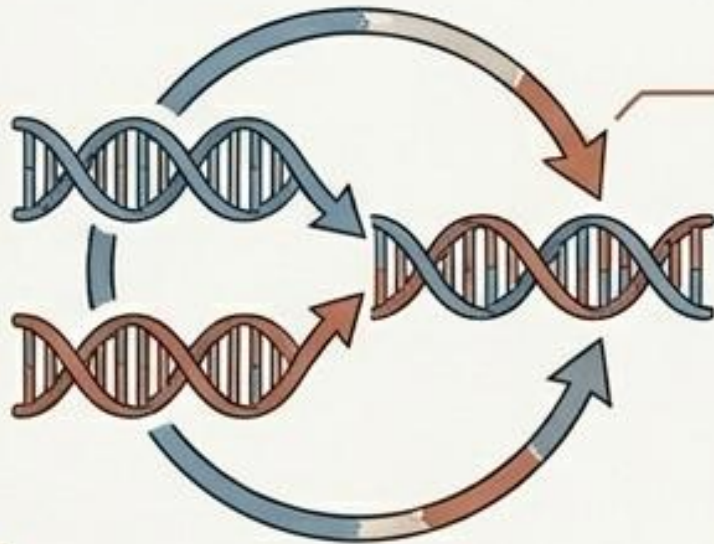
Sexual Strategy: r-Selected (Quantity)

Produces up to 100,000 offspring.
Zero parental care.
High mortality rate, relies on sheer numbers for survival.



Northern Pike.

Sexual Reproduction (General)



2 parents.
Offspring are genetically unique.
Energy-intensive but creates vital genetic diversity for adaptation.

Sexual Strategy: K-Selected (Quality)

Produces 1-2 offspring.
Massive energy invested in parental care.
High survival rate.



Black Bear.

ETHICS & DEFENSE: RESEARCH FRAMEWORKS & EVOLUTIONARY STRATEGIES

Model Organisms & The 3Rs



E. coli



Drosophila
(Fruit Fly)



Mouse

Ethical Research Framework: The 3Rs Principle



Replacement: Use non-animal alternatives if possible.



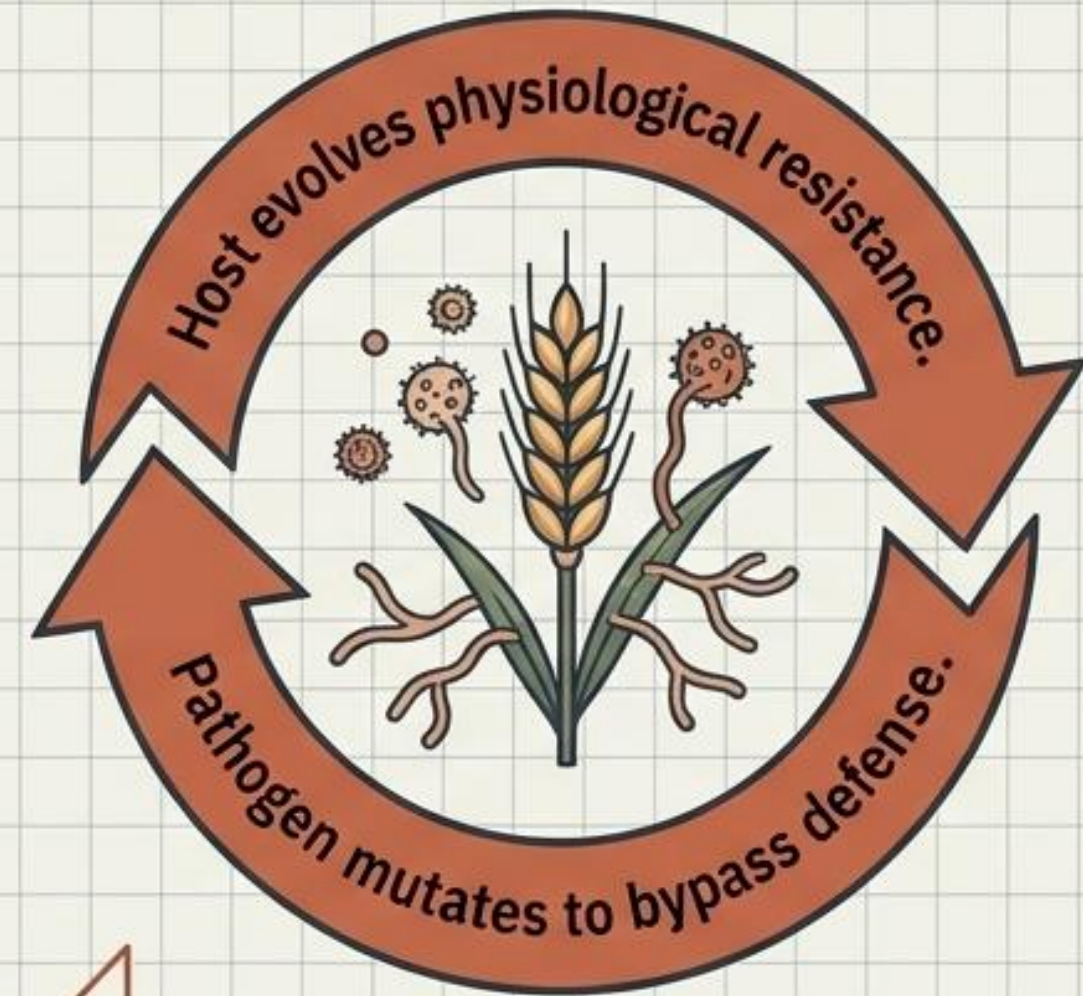
Reduction: Obtain valid data using fewer animals.



Refinement: Modify procedures to minimize distress.

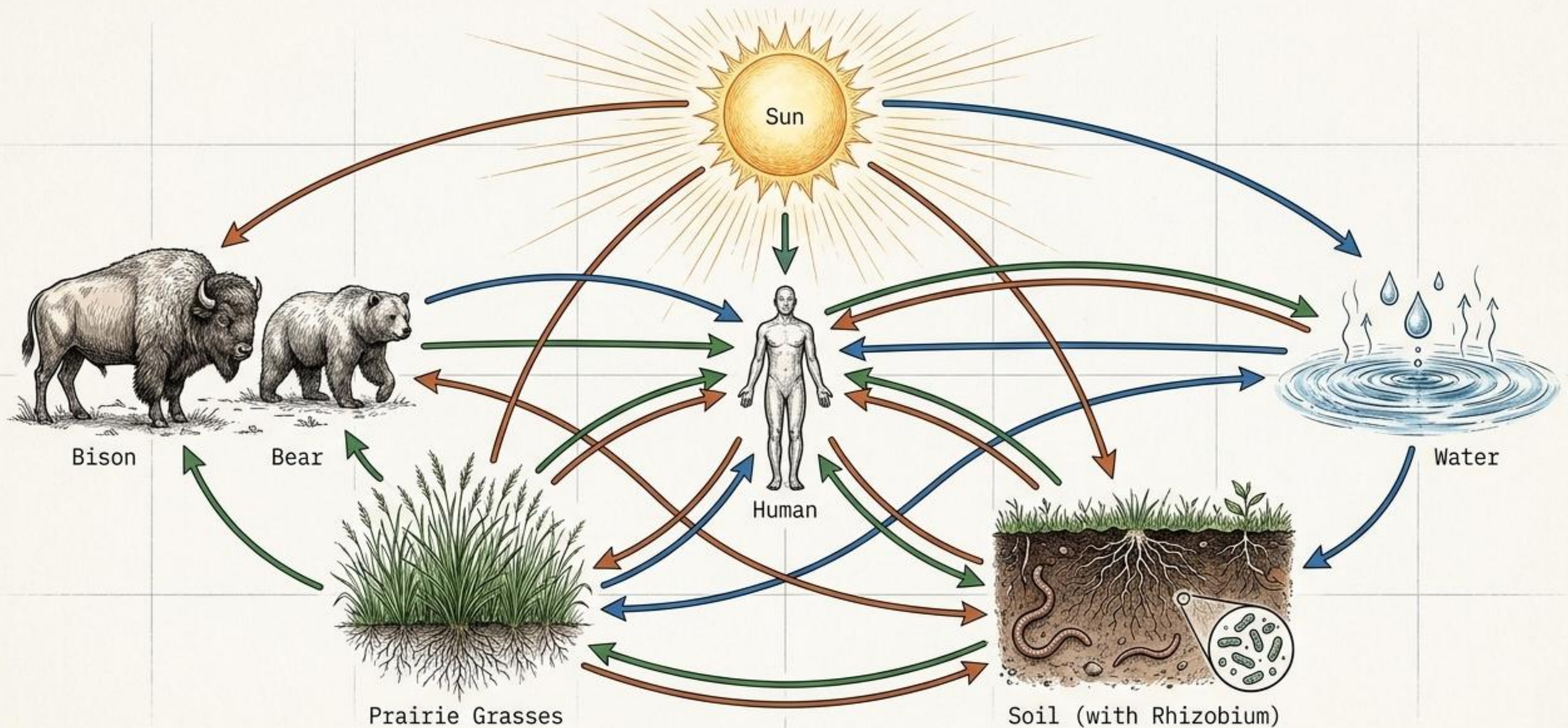
Defense: The Evolutionary Arms Race

Pathogen vs. Host



Saskatchewan Context: U of S researchers breed specific physiological resistance genes into wheat DNA to combat mutating Wheat Rust fungal pathogens.

Wâhkôhtowin: The Interconnected Architecture



From the folding of a single protein to the vast expanse of the living prairie, no organism exists in isolation. Life is the ultimate interconnected architecture.